

Chapter 6

Carbonyl Method of Metal Powder Production[☆]

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INTRODUCTION

The carbonyl method is one of the most widely used methods of chemical deposition [1–5]. Physical and chemical techniques of gas-phase manufacturing of metal powders have a common theoretical basis. A chemical reaction may answer for the formation of the metal vapor; its precipitation may occur because of simple condensation. At the same time, the evaporation-condensation processes also involve chemical interactions.

The process of chemical vapor deposition (CVD), mainly of high purity refractory metals, requires reagents able to vaporize without decomposition, form a sufficient concentration of the vapor of the metal-containing compound, and giving, upon decomposition of the latter and deposition of the metal, only gaseous and chemically inert substances.

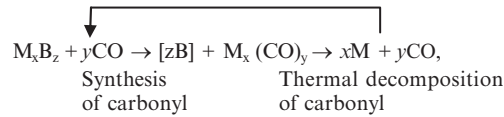
The decomposition of metal-containing compounds takes place as a result of a heat impact (plasma discharge, laser radiation). For chemical deposition, reducing agents are used, for example, hydrogen for deposition of tungsten from halides (see Chapter 23) and zinc or magnesium vapors for a quick deposition of titanium, zirconium, and beryllium from their halides. Using hydrogen lowers the process temperature. When a metal is used as a reducing agent, its quantity must strictly correspond to the stoichiometry of the reaction to prevent its coprecipitation and appearance in the powder. A lowered pressure serves the same result.

The method of chemical deposition from the gas phase is universal. It enables the manufacture of powders of a very high purity of the majority of refractory metals with targeted chemical composition, particle size, and shape; it also introduces necessary additives.

PREPARATION OF METAL CARBONYLS

The carbonyl method basis consists of the treatment of metal-containing materials with carbon monoxide at elevated pressures and temperatures followed by thermal decomposition of the metal carbonyls with deposition of the metal as a powder, and recycling of the carbon monoxide according to the scheme:

[☆] Expanded and updated by O.D. Neikov.



where B is the “ballast” matters (oxygen, salt residue, impurities, etc.); x , y , and z are the coefficients, the values of which depend on the nature of the metal; and $M_x(\text{CO})_y$ is the carbonyl of the metal M.

The flow sheet of the synthesis and decomposition of carbonyls is shown in Fig. 6.1.

The carbonyl process is used for manufacturing powders of nonferrous metals of VI, VII, and VIII groups of the periodic table (nickel, partly tungsten, molybdenum, cobalt, and some platinum group metals). Characteristic features of this process are low raw material and energy consumption, versatility of raw materials, high yield and purity of powders produced, and the possibility of completely automated operation. The technique affords control of particle size, shape, and structure. It makes possible the production of metallic films, coatings, filamentary crystals, and composite materials.

One of the shortcomings of the method is that it is applicable to only a limited range of metals. Other shortcomings include the toxicity of the materials, the high pressures involved in the process, the necessity of environmental control, and the health and safety measures.

Theoretical Basis of the Synthesis

Metal carbonyls are volatile and low-melting compounds of the $M_x(\text{CO})_y$ type that decompose upon heating into carbon monoxide and metal, their properties being conditioned by the structure of the carbonyl molecules and the type of chemical bonds in them. In the molecules of metal carbonyls, CO groups are bound to the atom of a metal via carbon atoms. The electron pair of the carbon atom is transferred to the metal atom with the formation of a σ -bond, and the α -electrons of the metal are transferred to vacant antibonding π^* -orbitals of CO (π -bond). The metal coordinates of such a number of CO groups as to complete its electron shell up to the one of the located behind it noble gas. It is assumed that CO donates two electrons, and the metal is in the zero oxidation state (in the neutral metal carbonyls).

The metal carbonyls may be mononuclear and polynuclear. In the polynuclear carbonyls, the metal atoms are bound to each other directly and in many cases also via bridging CO groups. The formula of carbonyls is determined according to the equations:

$$G - \frac{(xm + 2y)}{x} = x - 1 \quad (6.1)$$

and

$$y = \frac{x}{2}(G - m - x + 1) \quad (6.2)$$

where m is the metal atomic number; G is the atomic number of the nearest located behind noble gas; and x and y are the coefficients, the values of which depend on the nature of the metal.

Some properties of metal carbonyls are given in Table 6.1.

FIG. 6.1 The flow sheet of the carbonyl process.

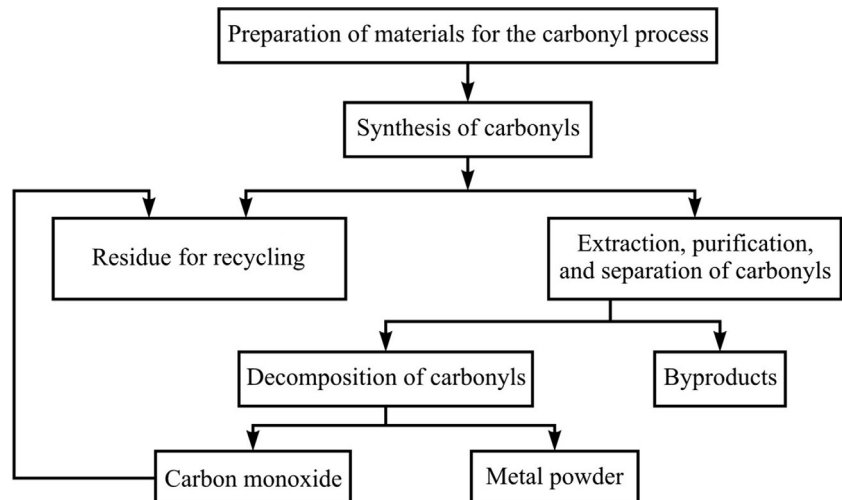


TABLE 6.1 Properties of Metal Carbonyls

Carbonyl	Physical State	Color	Temperature (K)				Density at 293 K (kg/m ³)	Reaction With Air	Solvents of Carbonyls
			Evaporation (sublimation)	Boiling	Melting	Decomposition			
Group V of Periodic Table									
V(CO) ₆	Solid	Blue-green	313	...	...	343	...	Yes	Ethers, pyridine, toluene, benzene
Group VI									
Cr(CO) ₆	Solid	Colorless	303	426	420	360–503	1.77	No	Chloroform, ethers, benzene, alcohols
Mo(CO) ₆	Solid	Colorless	313	428	421	403–673	1.96	No	Ethers, benzene
W(CO) ₆	Solid	Colorless	323	448	442	413–773	2.65	No	Chloroform, ethers, alcohols
Group VII									
Mn ₂ (CO) ₁₀	Solid	Golden-yellow	323	p*	427	383–573	1.81	Weak	Ethers, organic solvents
Tc ₂ (CO) ₁₀	Solid	Colorless	323	...	432	333–343	–	No	Ether, acetone
Group VIII									
Fe(CO) ₆	Liquid	Light-yellow	373	376	253 (freezing)	333–523	1.47	No	Petrol, benzene, ethers
So(CO) ₅	Liquid	Colorless	...	...	258 (freezing)	...	...-	–	Benzene (weakly)
Os ₃ (CO) ₁₂	Solid	Light-yellow	403	...	497	...	...	No	Benzene (weakly)
Co ₂ (CO) ₈	Solid	Orange	318	P	324	298–325	1.82	Yes	Petrol, benzene, alcohols
Co ₄ (CO) ₁₂	Solid	Dark green	...	P	...	333	...	Yes	Pentene, benzene
Rh ₂ (CO) ₈	Solid	Orange-yellow	–	P	349	...	...	Yes	Organic solvents
[Rh(CO) ₃] _x	Solid	Brown -red	...	...	...	473	...	No	Benzene (weakly)
Rh ₆ (CO) ₁₅	Solid	Black	...	...	...	493	...-	Yes	Organic solvents (weakly)
Ir ₂ (CO) ₈	Solid	Greenish-yellow	433	...	...	373–433	...	Yes	Ethers, CCl ₄
Ni(CO) ₄ ^a	Liquid	Colorless	303	316	248 (freezing)	333–473	1.31	No	Ethers, benzene, alcohols, acetone
[Pt(CO) ₂] _x	Solid	Dark red	...	P	...	483	3.55	Yes	Ketones, aniline, pyridine, alcohols

^aProperties of nickel carbonyl vapor: the density is 5.98 (in provided that air= 1.0) at 299K; the pressure is 400mm Hg at 280K.

p*—Decomposes below boiling point.

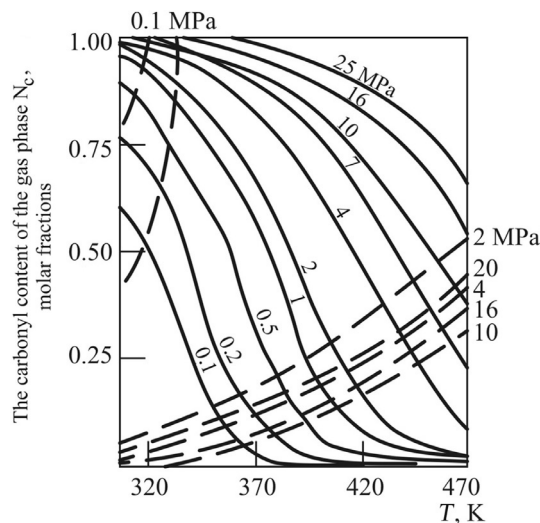


FIG. 6.2 The equilibrium content of nickel carbonyl in the gas phase as a function of the synthesis temperature under reaction (Eq. 6.3). Dashed lines divide the regions of condensed and gaseous state of the carbonyl.

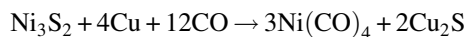
Metal carbonyls form upon a reaction of the metal in its active state with carbon monoxide, according to the reversible reaction:



The reaction is exothermic and is accompanied by a decrease in the number of moles of the products formed compared to the starting materials. Therefore, the process requires high pressures and moderate temperatures (Fig. 6.2).

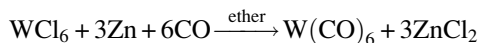
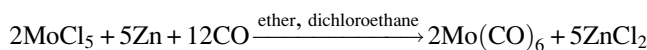
The synthesis of carbonyls takes place at the gas-solid boundary and goes through the following stages: carbon monoxide diffusion from the gas phase toward the solid surface that has cracks and pores; the formation of an absorbed layer of CO on the surface; the chemical reaction of CO with the metal in this layer with the initial formation of sub-carbonyls, followed by the formation of the $M_x(CO)_y$ molecule; and diffusion of metal carbonyl molecules from the solid surface into the gas phase. The reaction rate is, as a rule, limited by the diffusion of the gaseous compounds. Activation of the solid surface accelerates the synthesis reaction.

Metal carbonyls are obtained by a direct reaction of carbon monoxide with free metals, according to the reaction (Eq. 6.1) or by the reaction of carbon monoxide with metal compounds in the presence of reducing agents such as H_2 , Zn, Mg, Na, aluminum-organic compounds, and complex metal hydrides, for example:



The latter method is widely employed in practice due to available and low metal comprising raw materials.

Another known technique is carbonyl formation carried out in aqueous media or organic solvents:



In the process of synthesis of carbonyls of metals belonging to groups VI–VIII, a selective extraction of the metal from starting raw materials takes place while all the elements of the groups I–V except vanadium do not form carbonyls.

Preparation of the Starting Materials for Carbonyl Production

Carbonyls of cobalt and nickel are obtained either by direct reaction with pure metal or with metal-containing compounds. Thus, various oxidized and sulfide materials such as scrap metal, alloys based on Ni-Fe-Cu, Ni-Fe-Cu-S, and Ni-Cu-S systems, etc., are in use as starting materials for the commercial production of nickel carbonyl. To produce tungsten and molybdenum carbonyls, chlorides of these metals obtained from ore concentrates are used.

The reactivity of the starting materials depends on their structure, degree of defects, and chemical composition; it increases with the growth of surface area accessible for the chemical interaction. Therefore, a disintegration of the starting

materials to the optimal particle size in the range from 10 to 25 mm is preferable. Technological operations of crushing, granulating, or briquetting the starting materials, including the simultaneous addition of sulfur and oxidation, quenching, treatment with abrasives, etc., are used for this purpose.

Impurities can act either as inhibitors or as catalysts of the synthesis. Thus, copper, mercury, iron, sulfur, selenium, phosphorus, and some other elements increase the rate of carbonyl formation of nickel-containing materials while carbon, silicon, and chromium act conversely. The catalytic effect of impurities depends on their content and usually has a maximum, for example, for copper and for sulfur it equals $\approx 3\%$ and $3\text{--}8\%$, respectively.

In order to increase the metal content in the starting material and to intensify the reaction, raw materials are subjected to a pretreatment using the techniques of pyro- and hydrometallurgy, described in the chapters relating to the individual metals.

Synthesis of Carbonyls

The synthesis of most of carbonyls is carried out at carbon monoxide pressures in the range of 0.1–20 MPa and at temperatures of 320–550 K. The main parameters of the process are pressure, temperature, the flow rate of the gas stream, and the processing time. An increase in carbon monoxide pressure results in a higher stability of the carbonyls that allows the reaction temperature to increase and, accordingly, the process rate and the yield of the carbonyls (Fig. 6.2). However, the use of expensive high-pressure equipment involves an increase in both capital outlays and operational costs. The higher the reactivity of the starting material, the lower the optimal value of the synthesis pressure.

The yield of carbonyl depends on the temperature, growing with the rise in temperature up to the point where thermal decomposition of the carbonyl begins. A further rise of the temperature results in lowering the yield of the product. The intensity of heat and mass transfer in the reactor depends on the flow rate of the gas stream. Increasing the flow rate allows the use of a lower operating pressure of CO, or, at a given pressure, enlarges the output. The processing time determines the average rate of the process and the attainable extraction ratio of the metal.

The synthesis of carbonyls is performed in a hermetic vertical or horizontal reactor with cold walls cooled naturally or by external cooling means. The close contact and interaction of material with gas are attained by their countercurrent and the rotation of the reactor body or special mechanical devices. The reactors are usually provided with charging and unloading devices for preventing dust loss. There are continuous or batch reactors. The use of reactors with fluidized beds is quite promising (Chapter 21 contains a more detailed description of the reactor designs).

Carbonyl production in aqueous media is carried out in autoclaves equipped with stirring devices or in column reactors of a scrubber type. Pumps are used for the transportation of the reagents. The favorable features of these techniques are the continuity of the process and ease of control.

Extraction, Purification and Separation of Carbonyls

As a rule, the gas-vapor mixture of metal carbonyl with carbon monoxide coming out from a synthesis reactor contains admixtures including: carbonyls of elements relational to groups close to the base metal; ingress of water (up to 0.5%) out of the gas-holder; oil from compressors and pumps; and chlorine- and sulfur-containing compounds introduced by pretreatment of the raw materials. To eliminate the admixtures from the vapor-gas mixture, the carbonyls are liquefied by condensation at a pressure of 2–7 MPa. The purification of liquid carbonyls to a significant degree of purity is carried out in rectification columns by a conventional technology at atmospheric pressure. This technique is applicable to purification of nickel carbonyl containing 2%–3% max of iron carbonyl. Tungsten and molybdenum carbonyls are purified by steam distillation followed by alkaline washing to remove traces of chlorides.

If two or more carbonyls are formed in the synthesis process in comparable quantities, their purification and especially separation become more difficult. Thus, despite different volatilities of nickel, iron, and cobalt carbonyls, there are no means to prevent their decomposition when their mixture is separated at atmospheric pressure. The problem consists of sophisticated and expensive methods of separation such as rectification under pressure, fractional condensation, or some other special method. Therefore, in the commercial production of nickel, either starting materials with low iron content are used or mixed iron-nickel powders are produced. In processing tungsten-molybdenum raw materials where full separation of W(CO)_6 and Mo(CO)_6 is practically impossible, a preliminary separation of the metal chlorides is performed.

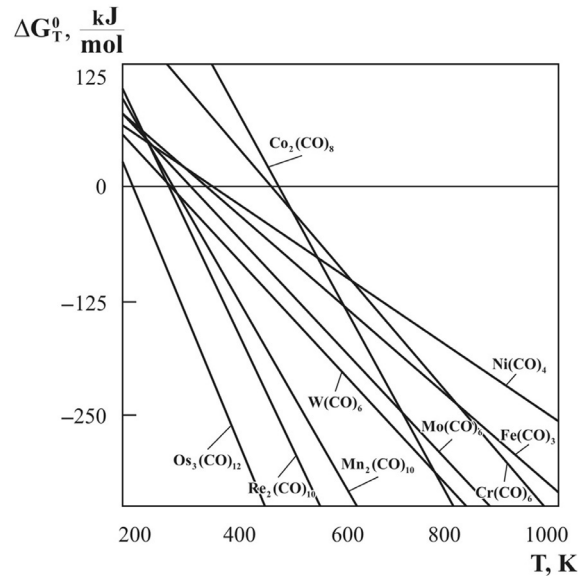


FIG. 6.3 Dependence ΔG_T^0 on temperature for the reaction of thermal decomposition of metal carbonyls.

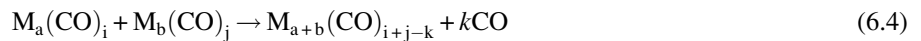
MANUFACTURE OF POWDERS BY THERMAL DECOMPOSITION OF METAL CARBONYLS

Physicochemical Basis of the Process

Thermal decomposition of metal carbonyls is an endothermic process, and according to the reaction (Eq. 6.3), it proceeds with the increase of the amount of the formed substance moles; therefore, the low pressures and high temperatures are expedient. According to the thermodynamic calculations (Fig. 6.3), trinuclear carbonyls are most prone and mononuclear ones are least prone to thermal decomposition.

The nucleation and growth of the new phase can proceed either heterogeneously on a prepared surface or in the gas volume. In the first case a metal coating and in the second case a metal powder arises.

The decomposition of the metal carbonyls proceeds stepwise, starting with the formation of unstable primary subcarbonyls and followed by splitting off the carbon monoxide molecules off to release the metal atoms. The latter by integration form nuclei of powder particles. The particles, the dimensions of which reach 4–20 nm, continue to grow due to thermal decomposition of the remained carbonyl vapor on their surface. At high concentrations of carbonyl in the gas phase, the fragments of the carbonyl can aggregate, and this process is accompanied by a partial splitting off of the CO group according to the scheme:



These aggregates act as the nuclei of the new phase. The process of nucleation depends on the kinetics of the reaction (Eq. 6.4), in the course of which the aggregates transform into metal particles with CO molecules sorbed onto their surface. Sintering and coagulation of the particles also contribute to their growth.

Side reactions of decomposition of the carbonyls lead to the formation of carbon, metal oxides, and carbides:



These reactions are typical for all the metals except the refractory metals.

The impurities influence the shape and structure of powder particles. The decomposition of CO according to (Eq. 6.5) results in a laminated “onion”-type structure of particles of carbonyl powders.

The carbon and oxygen content of the powders depends on the nature of the metal and on the process parameters. Side reactions have practically no influence on the properties of the metal powder. However, an exception must be made for metals, which do not form stable carbides, are slowly oxidized, and can affect as weak catalysts of the reaction of CO

disproportionation (for example, nickel). The influence of the side reactions is considerably higher for tungsten and molybdenum. Oxidation of the metal by carbon dioxide can take place when the latter is present in a considerable amount. The admixture of the iron carbonyl $\text{Fe}(\text{CO})_5$ in the amount of 1%–2% catalyzed the decomposition of the carbon monoxide.

The higher the process temperature, the lower the thermodynamic probability of the formation of carbides, oxides, and other admixtures. Therefore, the decomposition process should be carried out at a maximal allowable temperature to ensure the purity of the metals produced.

The Technique of Powder Manufacture

The decomposition of carbonyls is performed at carbon monoxide pressures close to atmospheric and at temperatures depending on the metal: 470–670 K for nickel, 620–1470 K for tungsten and molybdenum, and 420–520 K for cobalt. The reaction is carried out in a vertical column with external heating and filled with carbon monoxide. Carbonyl vapors from the evaporator are fed into the column. The greater part of the carbonyl decomposes in the upper section of the column; the growth and sintering of the particles formed take place in the middle and lower sections. The particles are initially suspended in the gas, and then precipitate and fall into a bunker. In some cases, forced circulation of carbon monoxide is applied.

The main process parameters are the temperature in the sections of the apparatus, the concentration of the carbonyl vapor at the entrance and its feed rate, the volume flow rate of the gases, and the presence of seeds. The temperature is the most important parameter controlling the powder particle shape and size. The higher the temperature in the top section of the apparatus, the more centers of nucleation of the metal arise, and the finer the size of powder particles.

Elevation the temperature from the top to the lower part of the apparatus intensifies the circulation of the reaction gas. The metal particles, which are being formed, repeatedly move with respect to apparatus height and particle residence time within the apparatus increases. The result is an increase of particle size.

Diluting the carbonyl vapor with an inert gas or carbon monoxide leads to the formation of finer powders and vice versa, increasing the vapor concentration leads to an increase of the particle size. The higher the gas volume flow rate, the shorter the residence time of the particles in the apparatus; accordingly, the finer powder is produced, and vice versa.

There are two basic modes of powder production in commercial usage:

- (1) A moderate rate of carbonyl is fed at a comparatively low temperature in order to stimulate the formation of a relatively small number of coarse primary particles and to obtain “heavy” powders (with high bulk density).
- (2) A high feed rate at a high temperature leading to the formation of a large number of small primary particles prone to coagulation into aggregates that results in the formation of “light” powders (with low bulk density).

In order to obtain fine and almost monodisperse powder, carbonyl vapors undergo decomposition by mixing immediately with a large volume of hot inert gas, followed by quenching the aerosol formed by diluting and cooling.

To increase the productivity of carbonyl decomposition, the nucleus seeds (metallic aerosol, solid particles) or appropriate chemical agents (oxygen, halogens, ammonia, HCl or HNO_3 vapors, chlorides or bromides of arsenic, boron, silicon, titanium, etc.) are introduced. The presence of these agents stimulates the transformation of carbonyls into oxides or other metal compounds used as crystallization nuclei.

The parameters of heat and mass transfer and, accordingly, the design of the apparatus influence the powder properties and process productivity. Heating or cooling of the apparatus sections as well as heating the introduced inert gas are used. To prevent metal deposition on the walls of the apparatus as well as to increase the process productivity, the gas flow is turbulent, stimulated either with a stirrer or with a tangential inlet of the gas. Adding to the height of the apparatus by lessening its diameter results in coarser powder. Placing a broad tube along the axis of the reactor decreases the free section and ensures the uniformity of powder particle size, etc. Feeding the carbonyl into the apparatus through spraying devices makes it more widely dispersed, the reaction volume is used more evenly, and the powder is more uniform.

The details of the carbonyl process for such metals as Ni, Co, W, and Mo are given in the corresponding sections.

A process, submitted in [6], comprises processing at atmospheric pressure and an effective temperature with hydrogen in the presence of chloride anion the nickel source including nickel chloride in admixture with nickel carbonate ore to produce resultant nickel and subsequent reacting the carbon monoxide with the resultant nickel to produce the $\text{Ni}(\text{CO})_4$.

A number of special methods for the decomposition of carbonyls are known. Ultraviolet and pulsed laser radiation, acoustic vibrations, and a magnetic field can be used for the manufacture of finely dispersed homogenous powders. In this process, a magnetic field below the Curie point results in a domain structure of the ferromagnetic particles formed.

In the plasma process, the reaction is carried out in highly ionized argon at a temperature of 4800–5300 K. Under those conditions, the number of nucleation centers increases sharply, and the dilution of the carbonyl vapors by the inert gas prevents coagulation of the particles. The result is the formation of submicron powders.

Manufacture of Composite Powders

The methods of producing composite powders by the carbonyl process are based either on the mutual solubility of metal carbonyls or on their ability to dissolve compounds prone to dissociation. Iron-nickel (permalloy) powders containing from 5% to 50% of nickel can be manufactured by simultaneous decomposition of a mixture of liquid iron and nickel carbonyls; the addition of cobalt and molybdenum is also possible.

Another method is based on the decomposition of a metal carbonyl on the surface of the introduced “seeds” (dispersed particles of metals, nonmetals, or oxides, carbides, sulfides, silicates of metals, and their mixtures). In this case, a metal film encapsulates both single particles and agglomerates of particles having different properties. The process is carried out in a fluidized bed. The intense mixing of the particles results in a uniform thickness of the metal coatings obtained. The deposition of a metal coating onto the particles imparts corrosion protection and electrical conductivity while increasing heat conductivity.

PROPERTIES AND APPLICATIONS OF THE METAL CARBONYL POWDERS

The size of carbonyl powder particles varies in a broad range from 0.5 to 20 μm . The features of these powders are high specific surface and considerable defects of crystalline structure; they are more active as compared to the powders manufactured by other techniques. However, these powders are not pyrophoric due to adsorbed carbon monoxide that passivates the particle surface. The particle shape is close to spherical. Sometimes particles have a laminated “onion” structure that imparts specific electromagnetic properties to the material. Carbonyl powders can contain 1–3% of carbon and oxygen (that is equivalent to about 30% of metallic compounds) while the amount of other impurities does not exceed $1 \times 10^{-3}\%$. Both typical spherically shaped nickel powders produced through a carbonyl process are shown in Fig. 6.4A with a spiky morphology and in Fig. 6.4B with a mottled surface.

The powders can be subjected to additional treatments with the purpose of bringing them into accordance with the requirements relating to the purity and the particle size. Mechanical methods (crushing, mixing, sieving, separating, and trapping) and thermal treatment are in use. The latter may cause changes in the crystalline structure of particles, thus enabling production of powders with specified properties, for example, light powders with coarse and porous particles. It can also cause an increase in the metal content of the powder up to 99.96% due to the elimination of carbon and oxygen.

Carbonyl powders of metals, alloys, and composites are used for manufacturing pure malleable metals, precision alloys, and various solid and porous items (parts, strips, rods, filters, etc.) as well as in electric and radio engineering, mechanical engineering, automation, and other areas. Detailed characteristics of powders are given for each metal in corresponding chapters of Section 4.

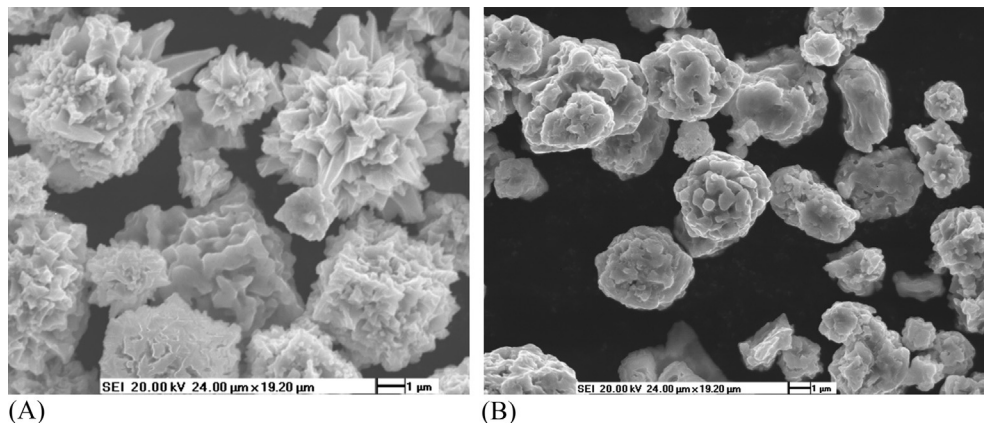


FIG. 6.4 SEM images of typical spherically shaped nickel powders produced by the carbonyl processes: (A) Nickel powder Grade AH50 with a spiky morphology, with Ni content 99.8% min and apparent density 2.1–2.4 g/cm³; (B) Nickel powder Grade AH410 with a mottled surface, with Ni content 99.8% min and apparent density 4.1–4.4 g/cm³. Courtesy: HUNTER Chemical LLC, PA, USA.

HEALTH AND ENVIRONMENTAL PROTECTION MEASURES IN THE CARBONYL INDUSTRY

The carbonyl process involves considerable danger such as increased flammability and explosiveness of the materials used, high toxicity, and high pressures employed in the process. Major health hazards are potentially fatal if inhaled, allergic reactions, cancer hazard. Physical hazards may include an explosion if exposed to shock, friction or heating; finely divided materials may explode spontaneously; extremely flammable liquid and vapor; vapor may cause a fire flash; liquid may decompose violently on heating; or possible decomposition on contact with air [7,8].

The mixture of nickel carbonyl and air or oxygen may ignite or explode. With its flammable and reactive nature, nickel carbonyl is classified as *serious* with a hazard rating of 3 by maximal ones 4 accordingly to the UA National Fire Protection Association (NFPA) [7–9]. Therefore, sources of ignition are prohibited while nonsparking equipment and tools and explosion-proof electrical equipment and fittings must be used. Metal containers involving the transfer of nickel carbonyl should be grounded and bonded.

The combustibility and explosive values of nickel carbonyl are characterized by the following: the flash point is 253 K; the autoignition temperature is 473 K, and the lower flammable limit in air is 2 vol%. The flammability class according to OSHA is 1B [8,9].

Toxic carbon monoxide can be responsible for acute and chronic poisoning in the manufacturing process. Under the legislation on ‘Health and Safety’ in the European PM industry the long- and short (15-min reference period)- term workplace exposure limits (WELs) for carbon monoxide in the workplace atmosphere are 35 mg/m³ and 232 mg/m³, respectively [10]. The long term WELs for carbon monoxide in workplace atmosphere is of 20 mg/m³ according to Legislation on ‘Health and Safety’ in the RUS Standard [11].

Metal carbonyls are toxic, with volatile ones such as nickel carbonyl being particularly dangerous. Nickel carbonyl is carcinogenic as well as flammable and reactive. Nickel carbonyl irritates and burns the skin and eyes. The WELs of the nickel carbonyl in the workplace atmosphere averaged over an eight-hour work shift under legislation in the United States, called *permissible exposure limits* (PEL), is 0.001 ppm. Exposure to 2 ppm (≈ 2.5 mg/m³) is immediately dangerous for life and health [8]. The WELs of the carbonyls in the workplace atmosphere under the CIS Standard are (in mg/m³): Ni(CO)₄, 0.0005 (≈ 0.0004 ppm); Co₂(CO)₈, 0.01; Mo(CO)₆, 1.0; and W(CO)₆, 2.0 [11].

The powders are also toxic, especially the nickel powder that affects the breathing organs; they are also carcinogenic. The long-term WELs of the powders in the air (mg/m³) are Ni, 0.05; Co, 0.5; Mo metal, 0.5; and W, 6.0 [11].

Implementation of enclosed operations, local exhaust ventilation, and general ventilation must be made to ensure the workplace exposure limit.

The industrial equipment for the carbonyl process must pass strict controls for gas tightness. All equipment and pipe work must be purged with an inert gas prior to feeding in the toxic and explosive compounds with the object of checking their gas-tightness as well as eliminating oxygen. When feeding in the carbon monoxide at operating pressures, control of its content in the air. The shop floors must be equipped with automatic analysis and alarm appliances for control of the presence of carbonyl vapors and carbon monoxide in the air. In all dangerous areas, personnel must wear protective breathing masks. Rubber-impregnated suits, rubber boots, gloves, and gas masks are a must for the personnel involved in handling liquid carbonyls.

The methods of determining small concentrations of carbonyls in the air are based on absorption, decomposition, or chemisorption of carbonyls by using various chemical reagents, combined with instrumental methods of analysis (colorimetric, gas chromatography, spectral analysis, etc.). The method for determining the carbonyl content in water is based on the volatility of carbonyls and consists in their being purged from the aqueous solution using nitrogen, followed by chemisorption from the gas phase using porous active materials and chemical reagents.

Liquid carbonyls must be poured only into special tanks under a water layer. Special gas-tight vessels must be used for the transportation of carbonyls and carbonyl-contaminated wastes to the place of disposal, according to local regulations. Solid wastes of carbonization fit for recycling and further processing undergo a pretreatment with oxidizing agents.

Nickel carbonyl must be stored to avoid heat, flames, sparks, and other sources of ignition. Nickel carbonyl is incompatible with halogens, oxidizing materials, and halocarbons. Therefore, avoid contact with oxidizing agents (such as perhalates, peroxides, permanganates, chlorates, nitrates, chlorine, bromine, and fluorine), strong acids (such as hydrochloric, sulfuric, and nitric), dinitrogen tetroxide, and tetrachloropropadiene because violent reactions may occur.

The major environment protection requirements relate to the technology for purification of the atmospheric ventilation discharges from toxic compounds and preventing the contamination of surface water according to Surface Water Standards & Guidance. The carbonyls can be present in considerable amounts in liquid waste due to their solubility in the water employed in the production process. Contaminated effluents, especially from the rectification unit, may have a dangerous impact on the human body, mainly via inhalation. The limits of the concentration of carbonyls in aqueous media must be evaluated in accordance with local environmental regulations.

Purification of ventilation discharges from carbonyls is complicated by their small concentration. The traditional way of air decontamination consists of thermal treatment or oxidation by using various agents (oxygen, chlorine, gaseous hydrogen chloride, phosphorus, etc.) on active porous materials (activated carbon, silicate absorbents, dry aluminum oxide, etc.). In a high frequency electric discharge, the molecules of carbonyls are destroyed and oxidized by ozone. The degree of air decontamination reaches 97%–99%.

The method of decontamination of carbonyl-containing water to a concentration below that of the admissible concentration limit [12,13] consists of purging the carbonyls by nitrogen, followed by passing water through the columns filled with activated carbon. The gas after purging is directed into the ventilation system and subjected to decontamination together with other gaseous discharges.

Health hazard information, including acute health effects, chronic health effects, cancer hazards, reproduction hazards, medical testing, mixed exposures, and conditions made worse by exposure, is contained in [9–11].

REFERENCES

- [1] Naboychenko SS, editor. Handbook of nonferrous metal powders. Moscow: Metallurgia Publishers; 1997 [in Russian].
- [2] Mnukhin AS, Shvartsman RA, Zaitsev AY. Synthesis of nickel tetra carbonyl at average pressure. *Tsvetnaya Metallurgiya Publ* 1998;2:64–6 [in Russian].
- [3] Syrkin VG. CVD-process. Moscow: Nauka; 2001 [in Russian].
- [4] Biketova LV, Kozyrev VF, Mnukhin AS. Carbonyl nickel powders for manufacturing of batteries. *Tsvetnaya Metallurgiya* 2001;2:73–5 [in Russian].
- [5] Lascelles K, Morgan LG, Nicholls D, Beyersmann D. Nickel Compounds. In: *Ullmann's Encyclopedia of Industrial Chemistry*. Wiley-VCH; 2005. p. 17–235.
- [6] Process for production nickel carbonyl, nickel powder and use thereof. Canadian Patent Application 2012, No: CA 2461624.
- [7] MATENSON TRI-GAS, Inc. Material safety data sheet. Nickel carbonyl. Revision date: Dec 11 2008. <https://www.mathesongas.com/pdfs/msds/MATH0059.pdf> [Accessed 31 January 2018].
- [8] US New Jersey Department of Health and Senior Services. Hazardous Substance Sheet. Nickel carbonyl. Revision: February 2001. <http://nj.gov/health/eoh/rtkweb/documents/fs/1343.pdf> [Accessed 31 January 2018].
- [9] UA National Fire Protection Association. Codes & Standards.
- [10] EH40-Update to EH40/2005 Workplace Exposure Limits. 2nd ed., Published 2011 (as consolidated with amendments, December 2011).
- [11] CIS Standard GOST 12.1.005–88.
- [12] Sanitary Regulations and Standards of surface waters from pollution. SanPiN 4630–88 (in Russian).
- [13] Surface Water Standards & Guidance. EPA United States Environmental Protection Agency. <https://www.epa.gov/wqs-tech/federal-water-quality-standards-requirements> May 16, 2017.